FOOT & ANKLE · NEET-SS / FET / INI-SS

Diabetic foot and Charcot

Part A: The diabetic foot

1. Epidemiology and pathogenesis

- **Lifetime risk of foot ulceration in diabetes: about 15-25%**; **about 85% of diabetic lower-limb amputations are preceded by an ulcer**; **5-year mortality after major amputation is about 50-70%** (worse than many cancers). **India has one of the largest diabetic populations** (barefoot walking, late presentation, infection).
- **Triad:**
  - **Peripheral neuropathy (commonest underlying cause of ulcers; about 80-90%)**:
  - **Sensory** (loss of protective sensation: **10-g Semmes-Weinstein monofilament**: **inability to feel it = loss of protective sensation**; **128 Hz tuning fork (vibration)**, **ankle reflexes**, **biothesiometer (VPT over 25 V)**),
  - **Motor** (intrinsic weakness: **claw toes, prominent metatarsal heads, fat pad displacement**: high plantar pressures),
  - **Autonomic** (**dry, cracked skin (anhidrosis)**, **arteriovenous shunting: warm foot with bounding pulses**, increased bone blood flow: osteopenia).
  - **Peripheral arterial disease (PAD)**: **infrapopliteal (tibial/peroneal) arteries**, **foot arteries often spared**; **medial arterial calcification (Mönckeberg sclerosis) makes ABI falsely high (over 1.3: non-compressible)**: **use toe pressures (over 30-40 mmHg needed for healing), toe-brachial index (TBI under 0.7 abnormal), transcutaneous oxygen pressure (TcPO2 over 30-40 mmHg predicts healing)**, **arterial Doppler waveforms**, **angiography**.

    - **Infection** (impaired immunity: neutrophil dysfunction with hyperglycaemia).
- **Other factors: deformity** (hallux valgus, claw toes, Charcot), **limited joint mobility (glycosylation of collagen: stiff joints, Achilles contracture: forefoot pressure)**, **previous ulcer/amputation (strongest predictor of new ulcer)**, poor glycaemic control, **renal failure (dialysis)**, smoking, poor footwear, visual impairment.

2. Classification of ulcers

Wagner (Meggitt-Wagner) classification

Grade	Description
0	Intact skin (pre-ulcerative lesion, deformity, healed ulcer)
1	Superficial ulcer (skin and subcutaneous)
2	Deep ulcer to tendon, capsule or bone (no abscess/osteomyelitis)
3	Deep ulcer with abscess, osteomyelitis or septic arthritis
4	Localised (forefoot/toe) gangrene
5	Whole-foot gangrene

- **Limitation: does not account for ischaemia or infection separately**.

University of Texas (UT) classification

- **Grade (depth): 0 pre/post-ulcerative, 1 superficial, 2 to tendon/capsule, 3 to bone/joint**, **Stage: A clean, B infected, C ischaemic, D infected and ischaemic** (**stage predicts amputation better than grade**).

Other

- **SINBAD** (Site, Ischaemia, Neuropathy, Bacterial infection, Area, Depth: **IWGDF recommended for communication**), **WIfI** (**Wound, Ischaemia, foot Infection**: **SVS, predicts amputation risk and benefit of**

revascularisation), **PEDIS** (IWGDF infection classification: perfusion, extent, depth, infection, sensation).

- **IWGDF/IDSA infection severity: uninfected**, **mild (local, cellulitis under 2 cm)**, **moderate (cellulitis over 2 cm or deeper tissue: abscess, osteomyelitis, septic arthritis, fasciitis)**, **severe (SIRS/systemic toxicity)**.

3. Assessment of the ulcer

- **Probe-to-bone test** (**sterile metal probe touching bone**): **high positive predictive value for osteomyelitis in high-prevalence settings** (and high negative predictive value in low-prevalence settings).
- **Radiographs** (**osteomyelitis changes appear after 10-14 days: periosteal reaction, cortical erosion, osteolysis**; gas in soft tissue; foreign bodies), **MRI (most sensitive and specific imaging for osteomyelitis: marrow oedema on T1 hypointense (confluent low T1 signal is the most specific), T2/STIR hyperintense; sinus tracts, abscesses)**, **labelled leucocyte scan/PET-CT** (when MRI contraindicated; helps differentiate Charcot from osteomyelitis).
- **Bone biopsy (histology + culture): gold standard for osteomyelitis diagnosis** (percutaneous through intact skin, or at surgery); **superficial swab cultures are unreliable** (deep tissue samples after debridement).
- **Inflammatory markers: ESR over 70 mm/h suggests osteomyelitis**; CRP, procalcitonin.
- **Organisms:**
  - **Mild/acute infections: aerobic gram-positive cocci (*S. aureus*, beta-haemolytic streptococci)**,
  - **Chronic/previously treated/severe: polymicrobial (gram-negatives: *Pseudomonas* (macerated wounds), Enterobacteriaceae; anaerobes (necrosis/ischaemia)), MRSA**.

4. Management principles

- **Multidisciplinary diabetic foot team** (**reduces major amputation rates by about 40-80%**): diabetologist, orthopaedic/vascular surgeon,

podiatrist, orthotist, wound nurse, microbiologist.

- **Offloading (the cornerstone for neuropathic plantar ulcers):**
  - **Total contact cast (TCC): gold standard** (**forces compliance: non-removable**; **contraindicated with infection, ischaemia, deep/draining wounds**; change weekly; **healing about 80-90% in 6-8 weeks**),
  - **Non-removable knee-high walker ("instant TCC")** (**equivalent**), removable walkers (compliance issues), **felted foam**, therapeutic footwear.
- **Debridement** (**sharp debridement of callus and necrotic tissue: converts a chronic wound to an acute one**), **moist wound healing**, **negative pressure wound therapy (NPWT: after debridement, post-amputation wounds)**, adjuncts (**hyperbaric oxygen (selected ischaemic wounds)**, **growth factors (becaplermin: PDGF)**, skin substitutes, **sucrose octasulfate dressings**).
- **Revascularisation** (**endovascular angioplasty or bypass**) for PAD before definitive surgery.
- **Infection:**
  - **Mild: oral antibiotics (covering gram-positive cocci) 1-2 weeks.**
  - **Moderate/severe: urgent surgical drainage/debridement (abscess, necrotising infection, gas, compartment involvement) + IV broad-spectrum antibiotics**; **"diabetic foot attack" = emergency**.
  - **Osteomyelitis: surgical resection of infected bone** (**± 1-3 weeks antibiotics after complete resection with clean margins**) or **prolonged antibiotics alone (6 weeks) in selected forefoot cases**.
- **Glycaemic control, nutrition, smoking cessation, renal optimisation.**

5. Surgical procedures

- **Prophylactic/corrective ("elective, curative, emergent" classification):**

  - **Achilles tendon lengthening (percutaneous Hoke)/gastrocnemius recession**: **forefoot plantar ulcers** (**reduces forefoot pressure: lowers recurrence (about 75% reduction at 2 years, Mueller)**: **transfer ulcers to the heel (overlengthening)**),
  - **Flexor tenotomy (percutaneous)** for **apical/tip ulcers of claw toes**,
  - **Metatarsal head resection (single or panmetatarsal)**, **sesamoidectomy**, **exostectomy** (Charcot rocker-bottom prominences), **hallux IPJ arthroplasty (Keller) for plantar hallux ulcers**.
- **Amputations (level by perfusion, function, soft-tissue coverage):**
  - **Toe/ray amputation**: **hallux amputation: transfer pressure to the lesser metatarsals**; **second toe amputation: hallux valgus**; **ray amputation (single lateral rays tolerated; first ray amputation: poor)**.
  - **Transmetatarsal amputation (TMA)**: good function; **balance with Achilles lengthening and tibialis anterior/peroneal considerations** (equinovarus deformity tendency).
  - **Lisfranc/Chopart amputations**: **equinus and varus deformity** (**tibialis anterior/posterior unopposed; Achilles unopposed**): **Achilles tenotomy + tibialis anterior transfer to the talar neck/lateral**.
  - **Syme amputation** (ankle disarticulation with **heel pad preserved and anchored to the tibia**; **heel pad migration**: commonest complication; end-bearing; needs a patent posterior tibial artery).
  - **Below-knee (transtibial) amputation (BKA)**: **preserve knee: energy expenditure about 10-40% above normal (BKA) vs about 60-100% (AKA)**; **long posterior myocutaneous flap (Burgess)**; **ideal tibial length about 12-15 cm below the joint line (about 2.5 cm per 30 cm of height)**; **fibula 1-2 cm shorter than the tibia**.
  - **Through-knee/above-knee**: non-ambulators, failed BKA, fixed knee contracture.
- **Minimum requirements for healing: ABI over about 0.45**, **toe pressures over about 30-40 mmHg**, **TcPO2 over 30-40 mmHg**, **serum**

albumin over 3.0 g/dL, **total lymphocyte count over 1,500/µL**.

6. Prevention

- **Annual foot screening for all diabetics** (**monofilament, vibration, pulses, deformity, skin, footwear**); **risk stratification** (IWGDF 0-3).
- **Education (daily self-inspection, never barefoot, moisturise, nail care)**, **therapeutic footwear/insoles** (**extra-depth shoes, custom insoles**: prevent recurrence), **podiatry (callus removal)**, **temperature monitoring (home: differences over 2.2 C between feet = impending ulcer or Charcot)**.

Part B: Charcot neuroarthropathy

7. Definition and pathogenesis

- **Progressive, non-infectious, destructive inflammatory process of bones and joints in a neuropathic foot**, leading to **fragmentation, dislocation, collapse and deformity**.
- **Causes: diabetes (commonest today, about 0.1-0.4% of diabetics; up to about 13% of high-risk)**, **leprosy (India)**, **syringomyelia (upper limb: shoulder/elbow)**, **tabes dorsalis (syphilis: knee, hip)** (historical commonest), **alcoholic neuropathy, spinal cord injury, congenital insensitivity to pain, myelomeningocele (ankle)**.
- **Theories:**
  - **Neurotraumatic (German) theory**: **repetitive unperceived microtrauma in an insensate foot**,
  - **Neurovascular (French) theory**: **autonomic neuropathy: hyperaemia (AV shunting), osteoclastic bone resorption, osteopenia**,
  - **Unifying modern theory: RANKL-OPG pathway**: **trauma triggers uncontrolled inflammation (TNF-alpha, IL-1beta, IL-6): increased RANKL, osteoclast activation**; **loss of protective pain response** allows continued loading.

8. Eichenholtz stages (natural history) (+ Shibata stage 0)

Stage	Name	Clinical	Radiographs
0 (Shibata, "prodromal/at-risk")	Inflammation	Warm, swollen, red foot after minor trauma	Normal (MRI: bone marrow oedema, microfractures)
I	Development (fragmentation)	Warm (over 2 C higher), swollen, erythematous; may be painless	Fragmentation, debris, joint subluxation/dislocation, osteopenia
II	Coalescence	Decreased warmth/swelling	Absorption of debris, sclerosis, early fusion of fragments
III	Reconstruction (consolidation)	No warmth; stable (possibly deformed) foot	Remodelling, bony ankylosis, deformity (rocker-bottom)

- **Each stage lasts months; the whole process about 1-2 years**.
- **Erythema that resolves on elevation (Brodsky elevation test: within about 1 minute)** suggests **Charcot rather than infection** (cellulitis does not resolve on elevation).

9. Anatomical classification (Brodsky; modified)

Type	Region	Frequency	Notes
1	Midfoot: tarsometatarsal (Lisfranc) and naviculocuneiform joints	Commonest (about 60-70%)	Rocker-bottom deformity with plantar ulceration (cuboid or medial cuneiform prominence); relatively stable
2	Hindfoot: subtalar, talonavicular, calcaneocuboid (Chopart)	About 10-30%	Instability; long time to consolidate

Type	Region	Frequency	Notes
3A	**Ankle (tibiotalar)**	**About 10%**	**Most unstable; varus/valgus; ulceration over malleoli; worst prognosis**
3B	**Calcaneal tuberosity (pathological avulsion fracture)**	Rare	**Loss of push-off, flatfoot**
(**4**)	**Multiple regions**		
(**5**)	**Forefoot (MTPJs)**		

- **Sanders and Frykberg classification (patterns I-V): I forefoot (MTPJ/IPJ), II tarsometatarsal (Lisfranc), III naviculocuneiform/TN/CC (Chopart), IV ankle/subtalar, V calcaneus.**
- **Rogers and Bevilacqua classification: anatomical location (1 forefoot, 2 midfoot, 3 hindfoot/ankle) + A no deformity, B deformity, C ulcer, D osteomyelitis.**

10. Diagnosis

- **Hot, swollen foot in a neuropathic diabetic: Charcot until proved otherwise** (often **misdiagnosed as cellulitis, gout, DVT or sprain**).
- **Temperature difference over 2 C** (infrared thermometry) vs contralateral foot.
- **Radiographs (weight-bearing): fragmentation, dislocation, rocker-bottom (loss of calcaneal pitch, negative lateral talo-first metatarsal angle), lateral column collapse.**
- **MRI** (stage 0: marrow oedema, microfractures).
- **Charcot vs osteomyelitis (key distinction):**

Feature	Charcot	Osteomyelitis
Location	**Midfoot (joints, periarticular)**, multiple joints	**Forefoot or calcaneus (adjacent to ulcers)**; usually single bone

Feature	Charcot	Osteomyelitis
Ulcer	**Often absent** (or over a bony prominence)	**Ulcer contiguous with the bone; probe to bone positive**
Erythema	**Resolves with elevation**	Persists
MRI	**Subchondral/periarticular oedema; preserved fat; "ghost sign" absent**	**Diffuse marrow replacement (low T1), sinus tract, abscess; "ghost sign" (bone disappears on T1 and reappears on post-contrast)**
Nuclear medicine	**Labelled leucocyte scan: mild uptake** (marrow)	**Intense uptake on labelled leucocyte scan** (combined with marrow scan)
Gold standard		**Bone biopsy**

11. Management

Acute phase (stage 0/I): non-operative is the mainstay

- **Immobilisation and offloading: total contact cast (TCC)** (**changed weekly initially, then every 1-2 weeks**; **non-weight bearing or protected weight bearing**) until **temperature difference falls under 2 C** (stage II/III: **about 3-6 months or longer**).
- **Then: Charcot restraint orthotic walker (CROW)** (**about 6-12 months**), then **custom footwear/AFO** (lifelong).
- **Bisphosphonates/calcitonin: no proven benefit on outcomes** (may reduce markers/temperature; not routine).
- **Monitor the contralateral foot** (**bilateral involvement in about 25-30%**).

Surgery

- **Indications: unstable/unbraceable deformity**, **recurrent/non-healing ulceration over bony prominences despite offloading**,

osteomyelitis, **progressive deformity**, **pain/instability (ankle Charcot)**.

- **Timing: traditionally in the coalescence/consolidation (stage II-III) phase**; **early surgery in stage I** increasingly accepted for unstable patterns (ankle) with **"superconstruct" fixation**.
- **Procedures:**
  - **Exostectomy** (**plantar bony prominence: medial column (medial cuneiform) prominences do better than lateral column (cuboid)**) for **stable deformities with ulcers**,
  - **Achilles lengthening/gastrocnemius recession** (reduces midfoot load; corrects equinus driving collapse),
  - **Reconstruction (realignment arthrodesis):**
  - **Midfoot: medial column beam (solid bolt through the first metatarsal into the talus), lateral column beams, plantar plating (tension side)**,
  - **Hindfoot/ankle: tibiotalocalcaneal (TTC) arthrodesis with a retrograde IM nail**, **circular (Ilizarov) external fixation** (**infection/osteomyelitis, poor bone, ulcers**),
  - **"Superconstruct" principles (Sammarco): extend fusion beyond the zone of injury to include unaffected joints**, **bony resection to shorten and decompress soft tissues**, **strongest device tolerated by soft tissue**, **devices applied in the position of maximum mechanical function (plantar plating)**,
  - **Two-stage management with infection** (**resection of infected bone, antibiotic spacer, external fixation, then definitive fusion**),
  - **Amputation (BKA)** for **recurrent ulceration/infection, failed reconstruction** (**can give better function than repeated salvage attempts**).
- **Complications: non-union (fibrous union may be stable and acceptable)**, **infection**, **hardware failure**, **recurrence**, **contralateral Charcot (from increased load)**, **amputation (about 5-15%)**, **mortality**.

Exam pearl: **Neuropathy (10-g monofilament) causes most ulcers; ABI falsely high in calcified vessels: use toe pressures/TBI/TcPO2. Wagner 0-5 (3 = abscess/osteomyelitis, 4 forefoot gangrene); UT grade + stage (stage predicts amputation). Probe-to-bone; MRI best imaging; bone biopsy gold standard.**

Exam pearl: **TCC gold standard offloading (not with infection/ischaemia). Achilles lengthening reduces forefoot ulcer recurrence (transfer heel ulcers). BKA energy cost much lower than AKA. Healing: ABI over 0.45, toe pressure over 30-40, TcPO2 over 30-40, albumin over 3.0, lymphocytes over 1,500.**

Exam pearl: **Charcot: hot swollen neuropathic foot, temp difference over 2 C, erythema resolves with elevation. Eichenholtz I fragmentation, II coalescence, III reconstruction (Shibata 0 = MRI oedema only). Midfoot (Lisfranc) commonest; ankle worst. Treat with TCC until temperature equalises, then CROW. Surgery: unstable/ulcer/infection; superconstruct; medial column beam; TTC nail/Ilizarov.**

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.